

Exam Date & Time: 08-Mar-2025 (02:15 PM - 03:45 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

APPLIED GRAPH THEORY [MAT 5301]

Marks: 30

Duration: 90 mins.

MULTIPLE CHOICE QUESTIONS

Answer all the questions.

Section Duration: 20 mins

Select the correct answer

- 1) If $(4, 3, 2, 2, x)$ is a degree sequence of a simple graph, then the value of x is (0.5)
- | | | | | | | | | | | | |
|----|---|--|----|---|--|----|---|--|----|---|--|
| 1) | 0 | | 2) | 1 | | 3) | 2 | | 4) | 3 | |
|----|---|--|----|---|--|----|---|--|----|---|--|
- 2) In a social network, if the sum of degrees of all users is 100, how many connections (edges) are there in the network? (0.5)
- 1) 50 2) 52 3) 54 4) 56
- 3) If G is a disconnected graph of order 101, then the minimum degree of G is (0.5)
- 1) less than 50 2) less than 51 3) less than 52 4) less than 53
- 4) In a self-complementary graph with 101 vertices, the number of edges is (0.5)
- 1) 2525 2) 2526 3) 2527 4) 2528
- 5) The number of cut vertices in a Hamiltonian graph of order 10 is (0.5)
- 1) 0 2) 2 3) 4 4) 6
- 6) The vertex independence number of a path graph with 15 vertices is (0.5)
- | | | | | | | | | | | | |
|----|---|--|----|---|--|----|---|--|----|---|--|
| 1) | 8 | | 2) | 6 | | 3) | 4 | | 4) | 2 | |
|----|---|--|----|---|--|----|---|--|----|---|--|
- 7) The vertex covering number of a cycle with 30 vertices is (0.5)
- 1) 14 2) 15 3) 16 4) 17
- 8) The number of cut edges in a tree with 24 vertices is (0.5)
- 1) 22 2) 23 3) 24 4) 25
- 9) If G is a connected graph with diameter 10, then diameter of the complement graph of G is at most (0.5)
- | | | | | | | | | | | | |
|----|---|--|----|---|--|----|---|--|----|---|--|
| 1) | 4 | | 2) | 3 | | 3) | 5 | | 4) | 6 | |
|----|---|--|----|---|--|----|---|--|----|---|--|

- 10) Maximum number of edges in a bipartite graph G with 14 vertices is (0.5)
- 1) 49 2) 51 3) 54 4) 58

Descriptive

Answer all the questions.

- 11) . (i) Prove that a connected graph on n vertices with $n - 1$ edges is a tree. (4)
- (ii) A tree T with 50 pendant vertices has an equal number of vertices of degree 2, 3, 4 and 5 and no vertices of degree greater than 5. Determine the order of T . (4)
- 12) Test whether the vector $(3, 3, 2, 2, 2, 2, 2)$ is graphical. If so, draw a simple graph with this vector as a degree vector. (3)
- 13) Show that a self-complementary graph is of order $4k$ or $4k+1$ for some integer k . Draw a self-complementary graph with 9 vertices. (3)
- 14) If G is Eulerian, then show that all its vertices have even degree. Give an example of a Hamiltonian graph that is not Eulerian. (3)
- 15) Let G be graph on n vertices, with vertex covering number α_0 and vertex independence number β_0 . Prove that $\alpha_0 + \beta_0 = n$. (3)
- 16) Prove that the chromatic polynomial $P_x(C_n, x)$ of a cycle C_n is $(x - 1)^n + (-1)^n(x - 1)$ for $n \geq 3$. (3)
- 17) Show that a bipartite graph G has no odd cycles. (2)
- 18) Draw a simple graph with vertex connectivity 2, edge connectivity 3 and minimum degree 4. (2)
- 19) Define a cut edge in a graph. What is the total number of cut vertices in a tree with 10 vertices and 6 pendant vertices. (2)

-----End-----

Event Name	Subject
MIT MPL - BTech-M Sc-MCA - II-IV and VI Semester - Midterm Examination - Mar 2025	MAT 5301:APPLIED GRAPH THEORY

#	Course Outcome	Description	Marks
1	1	CO 1	15
2	2	CO 2	15
Total			30

* NA - CO Not Selected

Event Name	Subject
MIT MPL - BTech-M Sc-MCA - II-IV and VI Semester - Midterm Examination - Mar 2025	MAT 5301:APPLIED GRAPH THEORY

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	2(6.67%)
3	Application	17	25(83.33%)
4	Analysis	1	3(10%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		19	30 (100%)